

Qissa — Product Requirements Document

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Overview

Qissa is a web app that writes a new picture-book story in seconds, with the child as the hero, in Urdu, Pashto and English. It is live at qissa-tau.vercel.app.

Problem. Children in Pakistan rarely find stories in the languages they hear at home, set in places they know. Parents who want a bedtime story that teaches kindness or eases a fear (the dark, a new school) have to invent one themselves.

Solution. A parent sets up the hero once: name, girl or boy, an animal avatar, language and story length. Each night the child picks a place and a lesson, and AI writes an illustrated, branching story they can read along with, followed by a short quiz.

Vision. Every child in Pakistan can be the hero of a story told in their own language.

Goals and non-goals

This release aims to make a new, personal, culturally familiar story a two-tap habit for families, at near-zero running cost.

Goals

- A story in under 15 seconds for short and medium lengths, with the child named as the hero.
- Full stories in Urdu, Pashto and English, switchable on any page.
- At most two questions per story (place, lesson); everything else set once in Settings.
- Safe for ages 4 to 8 by design, with no child data leaving the device by default.
- Always a story: if AI fails, an offline template story is served instead of an error.
- Run on free tiers (Vercel Hobby, Groq and Gemini free API keys).

Non-goals for this release

- User accounts, sign-in or syncing stories across devices.
- AI-generated illustrations (built but switched off).
- Payments or subscriptions.
- Native iOS or Android apps; the responsive web app serves phones.

Users and personas

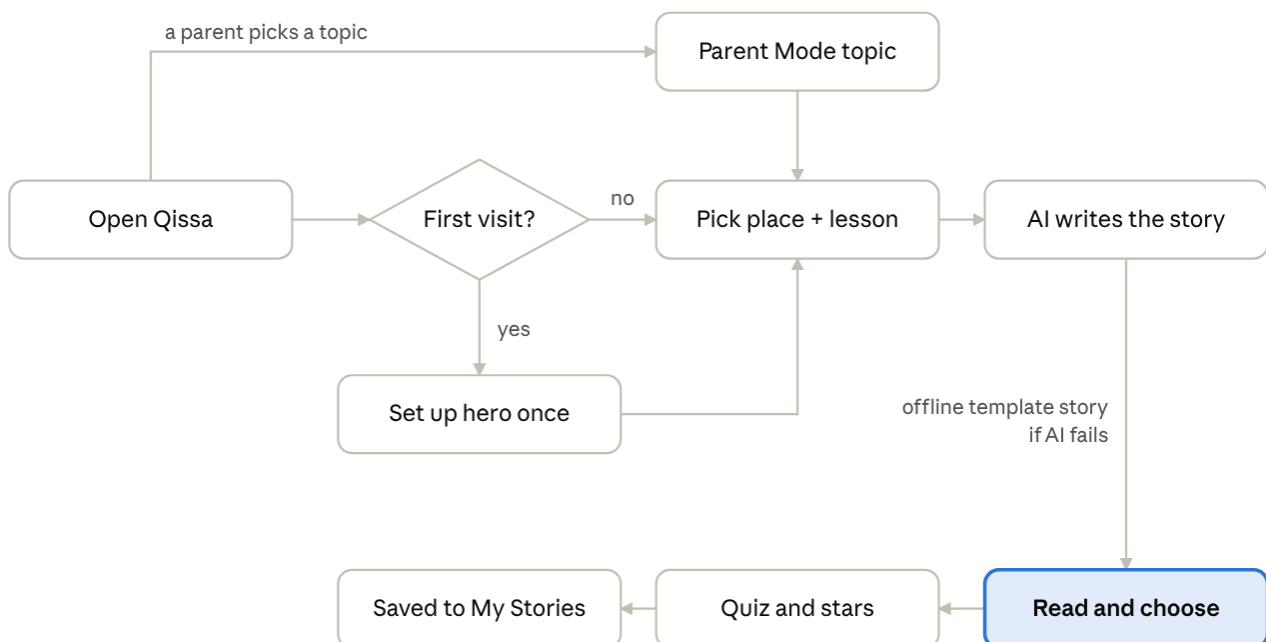
Qissa has two users at once: the parent who sets it up, and the child aged 4 to 8 who reads and plays.

Persona	Who	Needs	What Qissa gives them
Parent	Urdu- or Pashto-speaking parent in Pakistan or the diaspora, mostly on a phone	A quick, safe bedtime story; help with a big feeling	One-time setup, Parent Mode topics, no photos or sign-up
Child, 4 to 6	Pre-reader or early reader	To hear the story and choose what happens	Read-aloud with sentence highlighting, big tap targets, picture choices
Child, 7 to 8	Independent reader	To read alone and be challenged	Dual-language text, longer stories, quiz with stars

Core user journeys

After a one-time setup, every story takes two taps: a place and a lesson.

Two taps take a child from opening Qissa to a new story



user journey · setup once, then two taps per story

Parent Mode skips straight to the creator with a topic attached. "Change" on the creator reopens Settings, and any saved story reopens from My Stories into the reader.

Functional requirements

P0 items are needed for launch and all are built; P1 items are built but not yet working in production; P2 items are planned.

Area	Requirement	Priority	Status
Settings	Save hero name, girl/boy, animal avatar (8), language, story length, read-aloud options on the device	P0	Built
Settings	First visit opens setup before the first story	P0	Built
Story creator	One screen: pick a place (4) and a lesson (4); place pre-selected with the last one used	P0	Built
Story creator	Parent Mode topics: fear of the dark, new school, sharing, unkindness	P0	Built
Story	Short (6 pages, 3 quiz questions), medium (12, 4), long (20, 5)	P0	Built
Story	Branching choices that rejoin; every page reachable	P0	Built
Reader	Switch language on any page; dual-language view	P0	Built
Reader	Read-aloud with the current sentence highlighted; speed slow, gentle, normal; optional auto-read	P0	Built
Reader	AI voice for Urdu and Pashto when the browser has none	P1	Built, not working (needs Gemini key)
Quiz	Questions after the story with star rewards and a shareable result	P0	Built
My Stories	Stories saved on the device; reopen, favourite, delete	P0	Built
Illustrations	AI scene image per page	P2	Built, switched off

Area	Requirement	Priority	Status
Accounts	Sign-in and sync across devices	P2	Not started

AI story generation

A story is written in one request by a server function, checked against a fixed structure, and replaced by an offline template if anything fails.

Providers. Groq `openai/gpt-oss-120b` first, then `gpt-oss-20b` ; Gemini as a second provider once its key is added. The order is configurable (`AI_PROVIDER_ORDER`).

Output. Every title, page, choice and quiz item carries English, Urdu and Pashto text. The hero is written as a `{{HERO}}` token, so the child's name is filled in on the device.

Structure. Branching comes from one shared layout used by the AI prompt, the template stories and the reader. The prompt lists every page with its role, and the server sets the choice targets itself.

Length	Pages	Quiz	Decision points	Output budget	Time limit
Short	6	3	1 (page 1)	8,000 tokens	60 s
Medium	12	4	1 (page 4)	16,000 tokens	100 s
Long	20	5	2 (pages 6 and 16)	28,000 tokens	160 s

Reliability.

- One retry per model on rate limits, server errors and timeouts, then the next model.
- A wrong answer shape gets one corrective retry with the error spelled out.
- A Gemini model that fails twice in a row is skipped for 60 seconds.
- A hard deadline (time limit + 15 s) always returns the template story; no request can hang.
- Identical requests are cached for 5 minutes; 10 requests per minute per IP are allowed.

Child safety and privacy

Qissa collects no personal data on its servers: the child's name never leaves the device unless a parent turns on the AI voice.

Privacy

- The server receives only IDs: place, lesson, Parent Mode topic, language, length and girl/boy. The name is filled in on the device.

- No photos, accounts, emails or tracking. Avatars are cartoon animals.
- Settings and saved stories live in the browser's local storage only.
- AI voice is off by default; turning it on shows that the story text, including the name, is sent to a voice service.
- API keys live only in Vercel's encrypted environment variables, never in the browser.

Content safety

- The prompt requires gentle, age-appropriate stories rooted in Pakistani culture, with no violence, romance or frightening imagery.
- Gemini runs with its strictest safety settings.
- Every story is checked before it is shown: banned words (for example kill, weapon, gun), frightening themes (ghost, demon, horror, zombie) and over-long pages are rejected. "Scary" and "afraid" stay allowed so fear-of-the-dark stories work.
- A rejected story is retried or replaced by a reviewed template story.

Non-functional requirements

The app must feel instant on a mid-range phone and never leave a child waiting on an error.

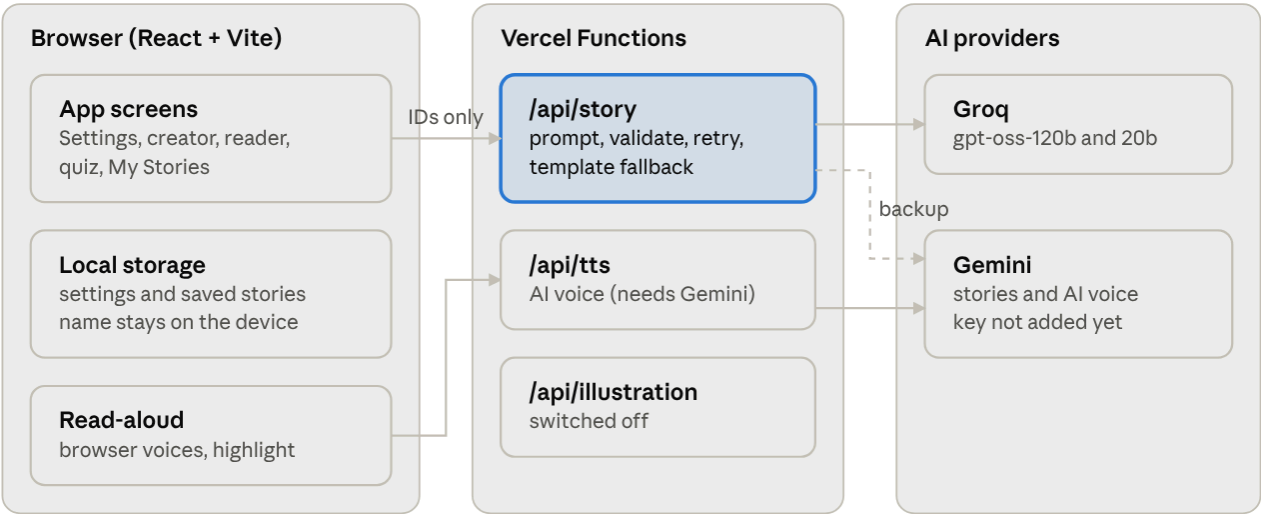
Area	Requirement	Measured so far
Speed	Short story ready in under 10 s; medium under 15 s	Short 3.8 to 4.2 s, medium 6.9 to 11.4 s, long 16 s (live tests, 3 Oct 2026)
Responsiveness	Every screen usable from 320 px phones to 1440 px desktops, no sideways scrolling	Checked at 320, 640, 768, 1024 and 1440 px
Accessibility	Every icon-only button has an accessible name; text readable for young readers	Labels added; story text enlarged on tablets and desktops
Reliability	Every request ends in a story within its time limit	Hard deadline and template fallback in place
Languages	Right-to-left layout and Nastaliq/Naskh fonts for Urdu and Pashto	Built
Cost	Run on free tiers at launch scale	Vercel Hobby plus free Groq key

Area	Requirement	Measured so far
Quality	Automated tests, typecheck and lint pass before each deploy	155 tests passing

Architecture and tech stack

Qissa is a React single-page app with three small Vercel Functions in front of the AI providers; there is no database.

Stories are written on the server; the child name stays in the browser



Code on GitHub; hosted on Vercel Hobby. Keys live only in Vercel environment variables.

architecture · browser, Vercel Functions, AI providers

Stack: React 18, TypeScript, Vite and Tailwind CSS in the browser; Node.js Vercel Functions; Zod for validating AI output; Vitest for tests. Supabase is installed but not used yet.

Success metrics

These are proposed targets for the first three months after launch; Qissa does not yet collect usage analytics, so measuring them needs privacy-safe analytics first.

Metric	Target	Why it matters
Setup completion: visitors who finish Settings	70%	Setup is the only form in the app

Metric	Target	Why it matters
AI success rate: stories served by AI, not the template	95% short and medium, 80% long	Template stories are generic
Time to story (median)	Under 8 s	A child is waiting on the loading screen
Stories finished: readers who reach the last page	60%	The story held their attention
Quiz completion	50% of finished stories	Learning is part of the promise
Repeat use: families creating 3+ stories in a week	30%	The bedtime habit

Current status, gaps and open questions

Qissa is live with AI stories from Groq; the latest responsive build is on GitHub but not yet deployed.

Known gaps

- ☐ Add the Gemini API key: needed for the AI voice and as a second story provider.
- ☐ Rewrite the AI voice request to match Gemini's text-to-speech API, then test Urdu and Pashto audio.
- ☐ Deploy the responsive build (commit cef583d) to production.
- ☐ Connect the GitHub repo to Vercel so every push deploys.
- ☐ Long stories on Groq's free tier often miscount pages or hit rate limits and fall back to the template.
- ☐ Rate limiting is counted per server instance; move it to Vercel Firewall before wide sharing.
- ☐ Pashto read-aloud works only with the AI voice, since browsers ship no Pashto voice.

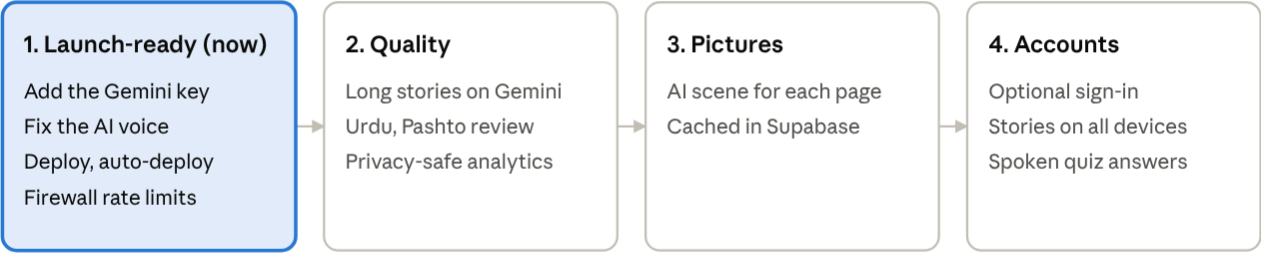
Open questions

- Should long stories stay available before Gemini is added, or be hidden until they work reliably?
- Which privacy-safe analytics will measure the success metrics?
- Who reviews AI story quality in Urdu and Pashto, and how often?

Roadmap

The next phase closes the launch gaps above; pictures and accounts follow once stories are reliable in all three lengths.

First make AI fully reliable, then add pictures, then accounts



Phases in order; dates not yet set.

roadmap · four phases, in order

Each phase starts when the previous one's items are working in production.